

Sarthak Shandilya

Data Scientist | Machine Learning Engineer

+91 9893846001 · sarthakshandilya9@gmail.com · [linkedin.com/in/sarthxk20](https://www.linkedin.com/in/sarthxk20) · github.com/sarthxk20 · sarthakportfolio.streamlit.app

SUMMARY

Data Scientist and ML Engineer with production experience across LLMops, generative modelling, predictive analytics, and time series forecasting — delivering ROUGE-1 0.50 (+138%), AUC ~0.97, ~60% forecast error reduction, and scalable end-to-end deployments via Docker, Kubernetes, and GitHub Actions CI/CD.

PROJECT EXPERIENCE

LLMOps Serving Platform — QLoRA Fine-Tuning + RAG Inference

Jun 2026 – Present

TinyLlama-1.1B · QLoRA · PEFT/TRL · FastAPI · FAISS · MLflow/DagsHub · Docker · Kubernetes · GitHub Actions
github.com/sarthxk20/llmops-platform · dagshub.com/sarthxk20/llmops-platform.mlflow

- Fine-tuned TinyLlama-1.1B-Chat on Bitext customer-support dataset using QLoRA (r=16, nf4 4-bit) via PEFT/TRL; achieved training loss 0.99→0.56, ROUGE-1 0.50 (+138%), ROUGE-2 0.27 (+200%), ROUGE-L 0.36 (+100%) across 3 epochs — tracked with MLflow on DagsHub (best run: ef3500f6507243efa16c76be568d7ab7).
- Built FastAPI inference server with FAISS-backed RAG using sentence-transformers; containerised via Docker multi-stage build with image pushed to GHCR; Kubernetes manifests (minikube) with namespace isolation; CI/CD via GitHub Actions — 27 pytest tests green, ruff lint clean, Docker image auto-published on every merge.

MorphGen — Conditional Generative Face Synthesis

Apr 2026 – Present

PyTorch · Conditional DCGAN · WGAN-GP · Spectral Normalisation · CelebA · Streamlit
morphgen.streamlit.app · github.com/sarthxk20/MorphGen

- Trained a Conditional DCGAN on CelebA (~200K images); applied WGAN-GP loss with gradient penalty and spectral normalisation — achieving attribute-conditioned face synthesis (gender, age, expression) across 50 epochs with stable convergence and no mode collapse.
- Engineered label conditioning via class embedding concatenation; resolved training instability by tuning critic-to-generator update ratio and adaptive LR scheduling; deployed interactive real-time inference on Streamlit.

PortIQ — GTM Opportunity Scoring Platform

Mar 2026 – Present

Random Forest · AIS Data · Scikit-learn · MLflow · Streamlit · SQL
portiq.streamlit.app · github.com/sarthxk20/PortIQ

- Conducted EDA and SQL-based querying on 1,200 AIS vessel records; engineered 27 domain-specific GTM features — Random Forest classifier achieved AUC ~0.97 with 5-fold CV across 180 companies, enabling precise prioritisation of high-value prospects.
- Automated tiered prospect ranking with TAM/SAM/SOM market sizing; tracked experiments via MLflow for reproducible feature engineering cycles.

DemandIQ — Retail Demand Forecasting & Risk Insights

Jan 2026 – Present

ARIMA · SARIMA · Prophet · XGBoost · PySpark · Docker · MLflow · Streamlit
demandiq.streamlit.app · github.com/sarthxk20/DemandIQ

- Forecasted demand across 1M+ retail records using ARIMA, SARIMA, Prophet, and XGBoost; reduced forecast error ~60% vs. naive baseline with Parquet-optimised PySpark pipelines and 14-day forward planning.
- Containerised CI/CD-ready pipeline with Docker and MLflow; delivered anomaly detection, scenario simulation, and confidence intervals via Streamlit for non-technical stakeholders.

CERTIFICATIONS

MLOps | Machine Learning Operations Specialization

May 2026

Duke University

MLOps Lifecycle, Model Deployment, CI/CD Pipelines, Model Monitoring, Containerisation, Kubernetes, Cloud ML

Machine Learning Specialization

Apr 2026

Stanford Online & DeepLearning.AI (Andrew Ng)

Supervised Learning, Unsupervised Learning, Reinforcement Learning, Neural Networks, Decision Trees

Machine Learning & AI with Python

Dec 2025

HarvardX (edX)

Ensemble Methods, Random Forest, Gradient Boosting, Neural Networks, Deep Learning, Bias-Variance Tradeoff

Data Science Professional Certificate

Oct 2025

IBM

Python, SQL, Machine Learning, APIs, Data Visualisation, End-to-End Model Development

Applied Data Science Lab

Jan 2026

WorldQuant University

Feature Engineering, EDA, Statistical Modelling, Applied ML, A/B Testing, Experiment Design, ETL, MongoDB

TECHNICAL SKILLS

Languages: Python, SQL

Data Engineering: Pandas, NumPy, SciPy, ETL, Parquet, REST APIs, MongoDB, Apache Spark, PySpark

ML & Statistics: Scikit-Learn, XGBoost, Random Forest, Regression, Classification, Clustering, Time Series, ARIMA, SARIMA, Prophet, Cross-Validation, Hyperparameter Tuning, A/B Testing, Feature Engineering

Deep Learning & GenAI: PyTorch, TensorFlow, Keras, Hugging Face Transformers, QLoRA, PEFT, TRL, LangChain, RAG, FAISS, sentence-transformers, OpenAI API, Whisper AI, LLMs, NLP, DCGAN, WGAN-GP, BERT, Prompt Engineering

MLOps & Deployment: MLflow, DagsHub, Docker, Kubernetes, GitHub Actions, CI/CD, FastAPI, Flask, AWS SageMaker, Azure ML, GHCR, Model Serving, Model Monitoring

Visualisation & Tools: Streamlit, Plotly, Seaborn, Matplotlib, Tableau, Power BI, Git, GitHub, Jupyter Notebook

EDUCATION

Bachelor of Arts

2020 – 2023

Dr. Harisingh Gour Central University — Sociology & History | CGPA: 8.15 / 10